WORKED SOLUTION

STEP III 2024, Question 1

Pure • Telescoping Series

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every sum here is handled the same way: split the term into partial fractions, pick out the pieces that are multiples of $1/r^2$ (these add up to a multiple of S_N), and arrange everything else into differences of the form $f(r) - f(r+1)$, which telescope. Work with finite sums up to N throughout, and only let $N \to \infty$ at the end, when every piece has a known limit.

Solution

Part (i)

Over the common denominator $r^2(r+1)$,

$$\frac{1}{r+1} - \frac{1}{r} + \frac{1}{r^2} = \frac{r^2 - r(r+1) + (r+1)}{r^2(r+1)} = \frac{r^2 - r^2 - r + r + 1}{r^2(r+1)} = \frac{1}{r^2(r+1)},$$

as required.

Now sum from $r = 1$ to N. The $1/r^2$ terms give S_N, and the other two terms telescope:

$$\sum_{r=1}^{N} \left(\frac{1}{r+1} - \frac{1}{r} \right) = \left(\tfrac{1}{2} - 1 \right) + \left(\tfrac{1}{3} - \tfrac{1}{2} \right) + \cdots + \left(\tfrac{1}{N+1} - \tfrac{1}{N} \right) = \frac{1}{N+1} - 1.$$

Hence

$$\sum_{r=1}^{N} \frac{1}{r^2(r+1)} = S_N - 1 + \frac{1}{N+1}.$$

As $N \to \infty$, $S_N \to \frac{1}{6}\pi^2$ and $\frac{1}{N+1} \to 0$, so the right-hand side tends to $\frac{1}{6}\pi^2 - 1$. The infinite sum is by definition the limit of the partial sums, so

$$\sum_{r=1}^{\infty} \frac{1}{r^2(r+1)} = \tfrac{1}{6}\pi^2 - 1.$$

Part (ii)

Write

$$\frac{1}{r^2(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r^2} + \frac{C}{r+1} + \frac{D}{r+2}.$$

Multiplying through by $r^2(r+1)(r+2)$ and using the cover-up rule: putting $r = 0$ gives $B = \frac{1}{2}$; $r = -1$ gives $C = \frac{1}{(-1)^2 \cdot 1} = 1$; $r = -2$ gives $D = \frac{1}{4 \cdot (-1)} = -\frac{1}{4}$. For large r the left-hand side is of order $1/r^4$, so the $1/r$ terms on the right must cancel: $A + C + D = 0$, giving $A = -\frac{3}{4}$. So

$$\frac{1}{r^2(r+1)(r+2)} = \frac{1}{2r^2} - \frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)}.$$

Ask yourself The three terms $-\frac{3}{4r}$, $\frac{1}{r+1}$, $-\frac{1}{4(r+2)}$ don't look like a single telescoping difference. How can we make them telescope?

Split the middle coefficient $1 = \frac{3}{4} + \frac{1}{4}$, so that each piece pairs with a neighbour:

$$-\frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)} = -\frac{3}{4}\left(\frac{1}{r} - \frac{1}{r+1}\right) + \frac{1}{4}\left(\frac{1}{r+1} - \frac{1}{r+2}\right).$$

Each bracket now telescopes:

$$\sum_{r=1}^{N}\left(\frac{1}{r} - \frac{1}{r+1}\right) = 1 - \frac{1}{N+1}, \qquad \sum_{r=1}^{N}\left(\frac{1}{r+1} - \frac{1}{r+2}\right) = \frac{1}{2} - \frac{1}{N+2}.$$

Therefore

$$\begin{aligned}\sum_{r=1}^{N}\frac{1}{r^2(r+1)(r+2)} &= \tfrac{1}{2}S_N - \tfrac{3}{4}\left(1 - \frac{1}{N+1}\right) + \tfrac{1}{4}\left(\frac{1}{2} - \frac{1}{N+2}\right) \\ &= \tfrac{1}{2}S_N - \frac{5}{8} + \frac{3}{4(N+1)} - \frac{1}{4(N+2)}.\end{aligned}$$

Letting $N \to \infty$, the last two terms tend to 0 and $S_N \to \frac{1}{6}\pi^2$, so

$$\sum_{r=1}^{\infty}\frac{1}{r^2(r+1)(r+2)} = \frac{\pi^2}{12} - \frac{5}{8}.$$

Part (iii)

The target has $\frac{2}{r^2(r+1)}$ on the right, so compare the two terms. Using part (i),

$$\frac{2}{r^2(r+1)} = \frac{2}{r+1} - \frac{2}{r} + \frac{2}{r^2}.$$

On the other hand,

$$\frac{1}{r^2(r+1)^2} = \left(\frac{1}{r} - \frac{1}{r+1}\right)^2 = \frac{1}{r^2} + \frac{1}{(r+1)^2} - \frac{2}{r(r+1)} = \frac{1}{r^2} + \frac{1}{(r+1)^2} - \frac{2}{r} + \frac{2}{r+1},$$

where we used $\frac{1}{r} - \frac{1}{r+1} = \frac{1}{r(r+1)}$ in both steps. Subtracting,

$$\frac{1}{r^2(r+1)^2} - \frac{2}{r^2(r+1)} = \frac{1}{(r+1)^2} - \frac{1}{r^2},$$

that is,

$$\frac{1}{r^2(r+1)^2} = \frac{2}{r^2(r+1)} - \left(\frac{1}{r^2} - \frac{1}{(r+1)^2}\right).$$

Ask yourself Why not just add infinite series term by term straight away?

Rearranging infinite sums is only safe once we know each piece converges, so we sum to N first. The bracket telescopes:

$$\sum_{r=1}^{N} \frac{1}{r^2(r+1)^2} = \sum_{r=1}^{N} \frac{2}{r^2(r+1)} - \left(1 - \frac{1}{(N+1)^2}\right).$$

By part (i), the first sum on the right converges (to $2(\frac{1}{6}\pi^2 - 1)$) as $N \to \infty$, and $\frac{1}{(N+1)^2} \to 0$. So the left-hand side converges too, and

$$\sum_{r=1}^{\infty} \frac{1}{r^2(r+1)^2} = \sum_{r=1}^{\infty} \frac{2}{r^2(r+1)} - 1,$$

as required. Both sides equal $\frac{1}{3}\pi^2 - 3$.

Checks

- Take $N = 1$ in part (ii): the sum is the single term $\frac{1}{6}$, and the formula gives $\frac{1}{2} - \frac{5}{8} + \frac{3}{8} - \frac{1}{12} = \frac{12-15+9-2}{24} = \frac{1}{6}$.
- Numerically, $\frac{1}{6}\pi^2 - 1 \approx 0.644934$, $\frac{1}{12}\pi^2 - \frac{5}{8} \approx 0.197467$ and $\frac{1}{3}\pi^2 - 3 \approx 0.289868$. Adding the first million terms of each series by computer reproduces all three values to at least nine decimal places.
- The finite identities in all three parts were confirmed exactly, with rational arithmetic, for every N up to 59.

Common slips

- Getting the end terms of a telescoping sum wrong, especially for the $\frac{1}{r+1} - \frac{1}{r+2}$ sum, which starts at $\frac{1}{2}$, not 1.
- Splitting into separate infinite series such as $\sum \frac{1}{r}$, which diverge, instead of keeping the harmonic-type pieces together in finite sums.

Beyond STEP

The value $\sum 1/r^2 = \pi^2/6$ is Euler's solution of the Basel problem, usually proved at university with Fourier series or with the product formula for $\sin x$. The care taken here over finite sums is exactly what analysis courses make precise: a series is the limit of its partial sums, and sums of convergent series may be combined, while divergent ones may not. Abbott's *Understanding Analysis* treats series this way and is a good next read.

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